
Module 01: Setting Up Your Dev Environment

Install and configure all the tools you need to build modern web applications with SvelteKit.

Revolution Trading Pros - Web Dev Course

Overview

Before writing any code, you need a properly configured development environment. In this module, we will install Node.js, pnpm (a fast package manager), Visual Studio Code (our code editor), and Git (for version control). Then we will create our first SvelteKit project and explore its structure.

By the end of this module, you will have a fully working development environment and a running SvelteKit application on your local machine.

Installing Node.js

Node.js is a JavaScript runtime that allows you to run JavaScript outside of the browser. SvelteKit requires Node.js version 18.13 or later. We recommend using the LTS (Long Term Support) version for stability.

macOS / Linux

The recommended way to install Node.js is using a version manager called nvm (Node Version Manager). This allows you to easily switch between Node versions for different projects.

Terminal

```
# Install nvm
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh |
bash

# Restart your terminal, then install Node.js LTS
nvm install --lts

# Verify installation
node --version
npm --version
```

You should see version numbers printed for both node and npm. If you see errors, make sure to restart your terminal after installing nvm.

Windows

On Windows, download the LTS installer directly from the official Node.js website at nodejs.org. Run the installer and follow the prompts. Make sure to check the box that adds Node to your PATH.

Terminal

```
● ● ●  
# After installation, open a new Command Prompt or PowerShell  
node --version  
npm --version
```

Verifying Your Installation

Run the following command to ensure everything is working correctly:

Terminal

```
● ● ●  
node -e "console.log('Node.js is working!')"
```

If you see the message printed, Node.js is installed and working correctly.

Installing pnpm

pnpm is a fast, disk-space-efficient package manager. It is significantly faster than npm and yarn because it uses a content-addressable store and hard links to avoid duplicating packages across projects.

Terminal

```
# Install pnpm globally
npm install -g pnpm

# Verify installation
pnpm --version
```

You should see a version number like 8.x.x or 9.x.x. pnpm is fully compatible with the npm ecosystem and reads the same package.json files.

Why pnpm Over npm?

- 🔧 Speed: pnpm installs packages up to 2x faster than npm.
- 🔧 Disk Space: pnpm stores packages in a global store and uses hard links, saving significant disk space.
- 🔧 Strictness: pnpm creates a non-flat node_modules structure, preventing phantom dependencies.
- 🔧 Monorepo Support: Built-in workspace support for managing multiple packages.

Installing Visual Studio Code

Visual Studio Code (VS Code) is a free, open-source code editor from Microsoft. It is the most popular editor for web development thanks to its excellent extension ecosystem, built-in terminal, and IntelliSense code completion.

Download and Install

Download VS Code from code.visualstudio.com and install it for your operating system. On macOS, drag the app to your Applications folder. On Windows, run the installer.

Essential Extensions

Install these extensions to supercharge your Svelte development:

- %İ Svelte for VS Code (svelte.svelte-vscode): Syntax highlighting, IntelliSense, and diagnostics for Svelte files.
- %İ Prettier (esbenp.prettier-vscode): Automatic code formatting. Set it as your default formatter.
- %İ ESLint (dbaeumer.vscode-eslint): JavaScript/TypeScript linting for catching errors early.
- %İ GitLens (eamodio.gitlens): Enhanced Git integration with blame annotations and history.
- %İ Auto Rename Tag (formulahendry.auto-rename-tag): Automatically renames paired HTML tags.

Terminal

```
# Install extensions from the command line
code --install-extension svelte.svelte-vscode
code --install-extension esbenp.prettier-vscode
code --install-extension dbaeumer.vscode-eslint
```

Recommended Settings

Add these settings to your VS Code configuration (Cmd/Ctrl + Shift + P, then 'Open Settings JSON'):

JSON

```
{
  "editor.formatOnSave": true,
  "editor.defaultFormatter": "esbenp.prettier-vscode",
  "editor.tabSize": 2,
  "editor.wordWrap": "on",
  "emmet.includeLanguages": { "svelte": "html" }
}
```

Git Basics

Git is a distributed version control system that tracks changes to your code over time. It allows you to collaborate with others, revert to previous states, and manage multiple features simultaneously through branching.

Installing Git

Git may already be installed on your system. Check by running:

Terminal

```
git --version
```

If it is not installed, download it from git-scm.com or install via your package manager:

Terminal

```
# macOS (via Homebrew)
brew install git

# Ubuntu/Debian
sudo apt install git

# Windows: Download from git-scm.com
```

Configuring Git

Set your name and email so Git can attribute your commits to you:

Terminal

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Essential Git Commands

Here are the Git commands you will use most frequently:



```
# Initialize a new repository
git init

# Check status of your files
git status

# Stage files for commit
git add .           # Stage all changes
git add filename.txt # Stage a specific file

# Commit staged changes
git commit -m "Your message"

# View commit history
git log --oneline
```

Think of Git like saving checkpoints in a video game. Each commit is a snapshot of your entire project at a specific point in time. If something breaks, you can always go back to a previous commit.

Creating a SvelteKit Project

Now for the exciting part! Let us create your first SvelteKit project. SvelteKit provides a CLI tool that scaffolds a new project with all the configuration you need.

Terminal

```
● ● ●  
# Create a new SvelteKit project  
pnpm dlx sv create my-first-app  
  
# Navigate into the project directory  
cd my-first-app  
  
# Install dependencies  
pnpm install
```

During the setup wizard, you will be asked several questions. For this course, select the following options:

% ☐ Template: SvelteKit minimal

% ☐ Type checking: TypeScript

% ☐ Additional options: Select Prettier and ESLint

The CLI will create a new directory with all the project files. Let us explore what was generated.

Project Structure Walkthrough

Understanding the project structure is crucial. Here is what each file and directory does:


```
my-first-app/  
% % % src/  
%   % % % routes/  
%   %   % % % +page.svelte      # Home page  
%   % % % lib/  
%   %   % % % index.ts         # Library exports  
%   % % % app.html              # HTML template  
% % % static/                   # Static assets  
% % % svelte.config.js          # Svelte configuration  
% % % vite.config.ts            # Vite bundler config  
% % % tsconfig.json             # TypeScript config  
% % % package.json              # Dependencies & scripts  
% % % pnpm-lock.yaml            # Lock file
```

Key Directories

- [%src/routes/](#): This is where your pages live. SvelteKit uses file-based routing, so the file structure maps directly to URL paths.
- [%src/lib/](#): Shared components, utilities, and modules. You can import from here using the `$lib` alias.
- [%static/](#): Files placed here are served as-is. Perfect for images, fonts, and other static assets.

Key Files

- [%app.html](#): The HTML shell that wraps your entire application. The `%sveltekit.head%` and `%sveltekit.body%` placeholders are replaced at build time.
 - [%svelte.config.js](#): Configures the Svelte compiler, adapter, and preprocessors.
 - [%vite.config.ts](#): Configures Vite, the blazing-fast build tool that powers SvelteKit.
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Running the Dev Server

Start the development server with hot module replacement:

Terminal

```
pnpm dev

# Output:
# VITE v5.x.x ready in 500 ms
#
# 'æ Local:   http://localhost:5173/
# 'æ Network: use --host to expose
```

Open your browser and navigate to <http://localhost:5173>. You should see the SvelteKit welcome page. Try editing `src/routes/+page.svelte` - the browser will update instantly without a full reload.

Understanding Hot Module Replacement (HMR)

HMR is a development feature that updates your application in the browser without losing state. When you save a file, Vite detects the change, recompiles only the affected module, and sends the update to the browser. This makes development incredibly fast and productive.

Your First Edit

Open `src/routes/+page.svelte` in VS Code and replace its contents:

Svelte

```
<h1>Hello from SvelteKit!</h1>
<p>I just set up my development environment.</p>

<style>
  h1 {
    color: #ff3e00;
    font-size: 2rem;
  }
  p {
    color: #666;
  }
</style>
```

Save the file and check your browser. You should see your changes reflected immediately. Congratulations - you just wrote your first Svelte code!

Module 1 Exercise

Exercise: Set Up and Customize Your Project

Put what you have learned into practice with this hands-on exercise.

- 📌 Step 1: Create a new SvelteKit project using `pnpm dlx sv create my-portfolio`.
- 📌 Step 2: Install dependencies with `pnpm install`.
- 📌 Step 3: Open the project in VS Code.
- 📌 Step 4: Edit `src/routes/+page.svelte` to display your name, a short bio, and your favorite programming language.
- 📌 Step 5: Style the page with custom colors and fonts using a `<style>` block.
- 📌 Step 6: Initialize a Git repository with `git init`, stage all files with `git add .`, and create your first commit.
- 📌 Step 7: Take a screenshot of your running application and your terminal output to share with the community.

Summary

- 📌 Node.js provides the JavaScript runtime needed for SvelteKit development.
- 📌 pnpm is a fast, efficient package manager that we will use throughout the course.
- 📌 VS Code with the Svelte extension provides an excellent development experience.
- 📌 Git tracks your code changes and enables collaboration.
- 📌 SvelteKit projects are created with `pnpm dlx sv create` and run with `pnpm dev`.

In the next module, we will dive into HTML and CSS fundamentals within the context of Svelte components.